Spectrophotometry and Beer's Law

Absorbance, Beer's law and calibration curves

Name _____ Class _____ Date _____

EQUATIONS AND DATA

$T = I \div I_0$ · $\%T = T \times 100\ \%$ · $A = -\log T$ · $\%T = 10^{\wedge}(-A) \times 100\ \%$

Beer's law: $A = \varepsilon b c$ (ε in L mol^{-1} cm^{-1}, b in cm, c in mol/L)

Mean and sample standard deviation $(n - 1)$ · $RSD = s \div \text{mean} \times 100\ \%$ · $Cu = 63.55$ g/mol

Absorbance and transmittance have no units; show units on every other quantity.

A. Understanding

1. (a) Define transmittance and absorbance. (b) Why are measurements made at λ_{max}? (c) What is a blank, and why is it needed?

B. Absorbance and transmittance

2. Find the absorbance for: (a) T = 0.500 (b) T = 0.100 (c) 35.0 %T. Find the %T for: (d) A = 0.200 (e) A = 1.00

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C. Beer's law

3. A solution in a 1.00 cm cell has A = 0.725. The molar absorptivity is 1.45×10^4 L mol^{-1} cm^{-1}. Find the concentration.

4. A 2.50×10^{-5} mol/L solution gives A = 0.360 in a 1.00 cm cell. Calculate the molar absorptivity, with its units.

5. For the solution in question 4, predict the absorbance (a) in a 2.00 cm cell (b) at half the concentration in a 1.00 cm cell. Explain using Beer's law.

6. An unknown gives A = 1.85, which is above the calibration range. 10.00 mL of it is diluted to 50.00 mL, and the diluted solution gives A = 0.370 in a 1.00 cm cell ($\varepsilon = 1.20 \times 10^4$ L mol^{-1} cm^{-1}). Find the concentration of (a) the diluted solution (b) the original unknown.

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D. Calibration and replicates

7. A set of standards gives the best-fit line $A = (0.0820\ \text{L/mg})\,c + 0.003$. An unknown is measured in triplicate: A = 0.334, 0.338, 0.331. (a) Find the mean absorbance, the standard deviation and the RSD. (b) Find the concentration of the unknown.

8. Explain (a) why the unknown in question 7 is measured three times instead of once (b) why the unknown should lie inside the range of the standards (c) what a large intercept on the calibration line might show.

E. Challenge

9. 0.1000 g of brass is dissolved in acid, the copper is converted into a coloured complex, and the solution is diluted to exactly 1.000 L. It gives A = 0.409 in a 1.00 cm cell, and ε of the complex is 395.0 L mol^{-1} cm^{-1}. Calculate the percentage by mass of copper in the brass.

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